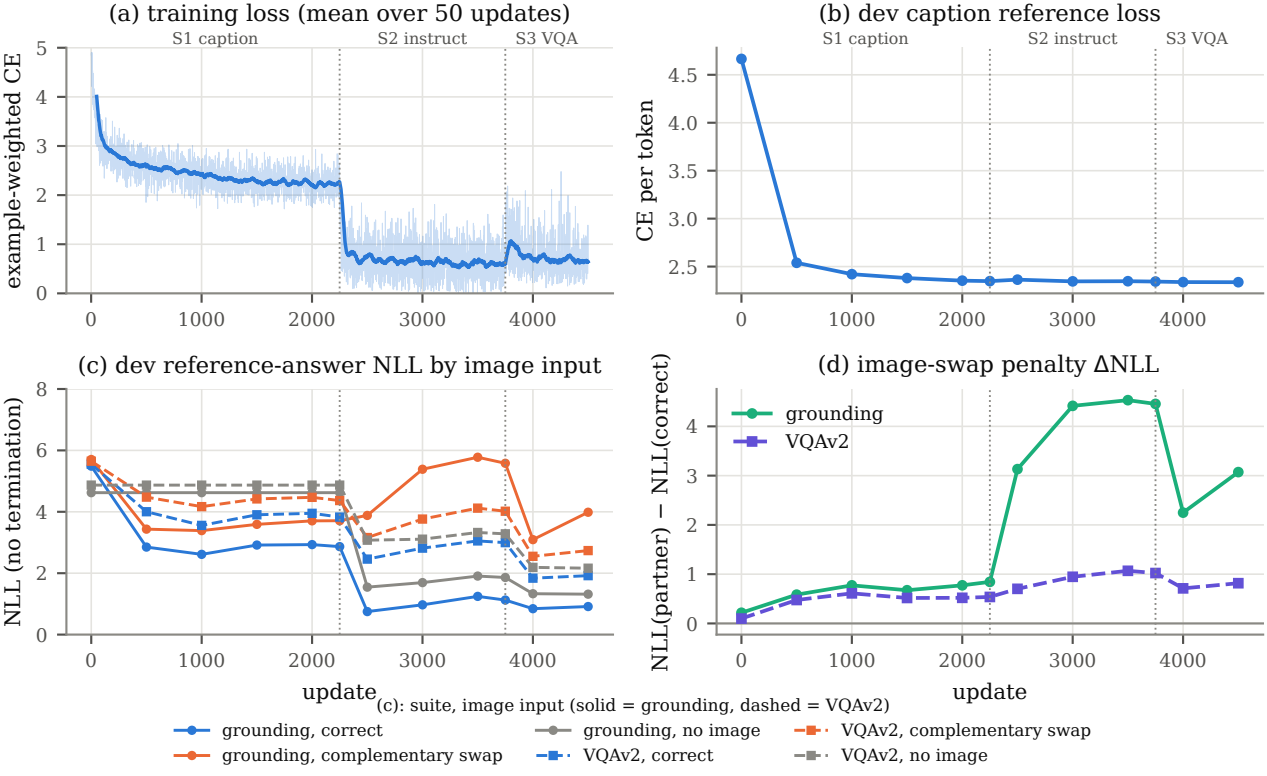

(a) training loss (mean over 50 updates)
S1 caption
S2 instruct
S3 VQA
example-weighted CE
(b) dev caption reference loss
CE per token
(c) dev reference-answer NLL by image input
NLL (no termination)
update
(d) image-swap penalty ΔNLL
grounding
VQAv2
NLL(partner) − NLL(correct)
(c): suite, image input (solid = grounding, dashed = VQAv2)
grounding, correct
grounding, complementary swap
grounding, no image
VQAv2, correct
VQAv2, complementary swap
VQAv2, no image